Understanding Host–Pathogen Epidemiology and Coevolution through joint Coalescent Genealogies

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Introduction

Coevolutionary theory

Models of reciprocal coevolution between hosts and pathogens, include frequency-dependent selection and interaction matrices of host and pathogen genotypes, leading to coevolutionary cycling or arms races [1].

While genomic signatures of coevolution have been summarized with allele frequencies and correlations [2], there has been *limited genealogical framing of host-pathogen interactions in context of coalescent trees*, limiting our ability to link genomic data to models.

Phylodynamics

Epidemiological dynamics of rapidly evolving pathogens can be generalized as coalescent processes [3]. This provides a theoretical framework linking genealogical trees of sampled pathogens to underlying population dynamic processes and thus powerful methods for studying infectious disease outbreaks. Crucially, phylodynamics assumes host evolution is negligible on the timescale of pathogen transmission.

Why joint genealogies?

It is not well understood how coevolution shapes the joint relationship of host and pathogen genealogical trees as determined by their respective infection and reproduction processes, across both short epidemic and longer coevolutionary time scales. More specifically, for infectious diseases the pathogen ancestral lineages are *embedded* within a host genealogy, mediated by horizontal and vertical transmission.

We present a neutral model of joint host–pathogen reproduction and transmission and obtain insights into their joint genealogies using both analytical methods and forward-in time simulations of the coalescent trees using *SLiM* [4].

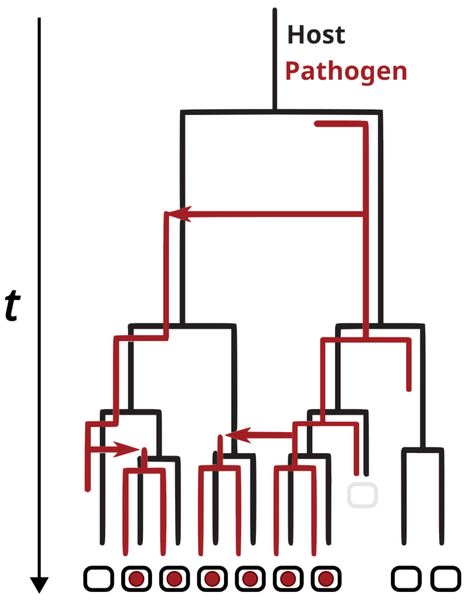
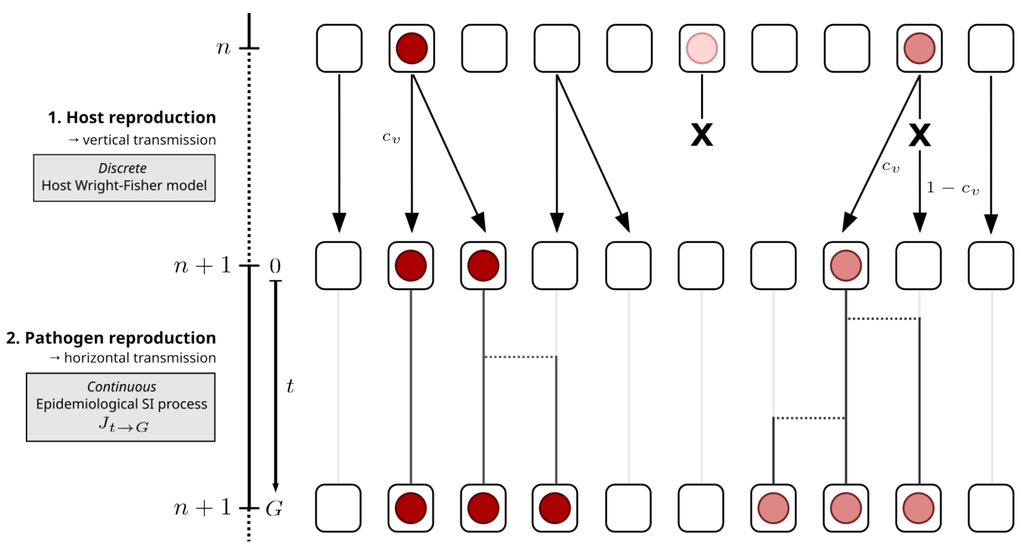


Figure 1: Pathogen genealogical tree embedded in a host genealogy through horizontal and vertical transmission (adapted from [5])

A nested semi-discrete Host-Pathogen model with vertical and horizontal transmission



Vertical transmission. The host population reproduces in discrete and non-overlapping generations, following a Wright-Fisher process of size N, with X_n infected indivduals at the end of generation n. Infected host can produce infected offspring through vertical transmission with probability c_v. At the start of the next generation are I_{n+1} initally host infected offspring:

$$I_{n+1} \sim \text{Binom}(N, c_v X_n / N) \tag{1}$$

Horizontal transmission. Continuous-time epidemiological infection process within a host generation running for time G and dynamics following an SI model, with intital conditions $\mathcal{J}_0 = I_{n+1}$ and the host contact rate β.

$$\mathcal{J}_t : \quad J_0 \to J_G \quad \text{at rate } \beta J_t (N - J_t)/N. \tag{2}$$

Combined process. The number of infected X_{n+1} at the end of each generation $n+1$ is the combination of vertical and horizontal infections.

Figure 2: Semi-discrete host-pathogen model with vertical transmission following host reproduction and horizontal epidemiological process within host generations.

Deterministic equilibrium and backwards process

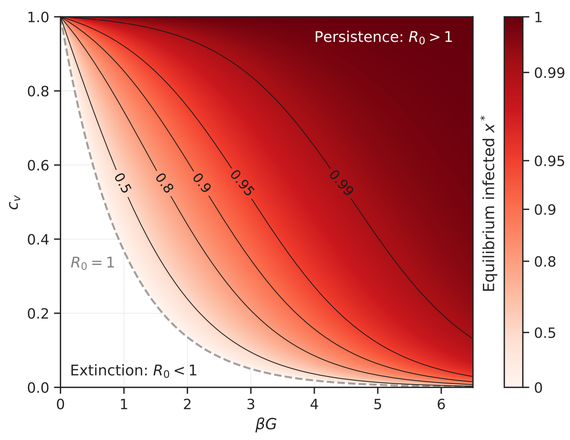


Figure 3: Equilibrium fraction of infected x^* and persistence threshold for rates of vertical transmission c_v and epidemic transmission β and epidemic time G.

Deterministic limit. Taking $N \to \infty$ and $x = X/N$ we get the following deterministic solution.

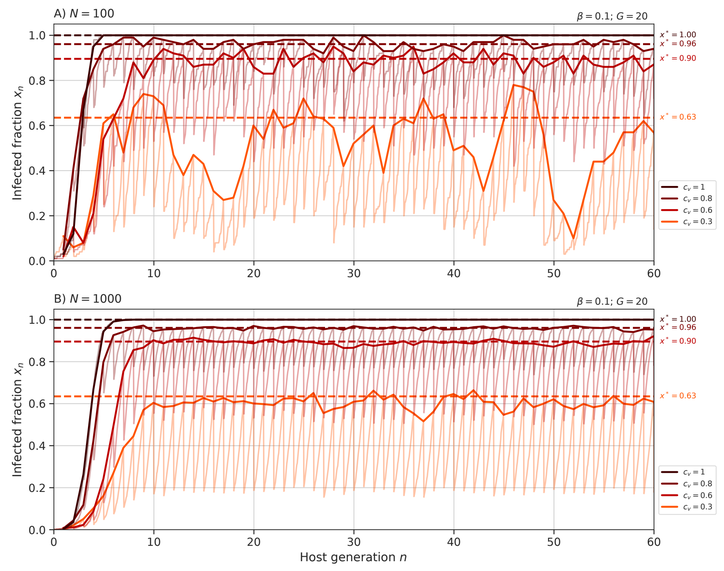
$$x^* = \frac{c_v - e^{-\beta G}}{c_v(1 - e^{-\beta G})}. \tag{3}$$

Basic reproductive number: $R_0 = c_v e^{\beta G}$ The persistence threshold for the pathogen ($R_0 > 1$) follows for $c_v > e^{-\beta G}$.

Ancestral process. The probability for two randomly sampled pathogen lineages to coalesce within a host generation:

$$p_c = \frac{1}{N} + \frac{2(1 - c_v)}{c_v x^* N} \tag{4}$$

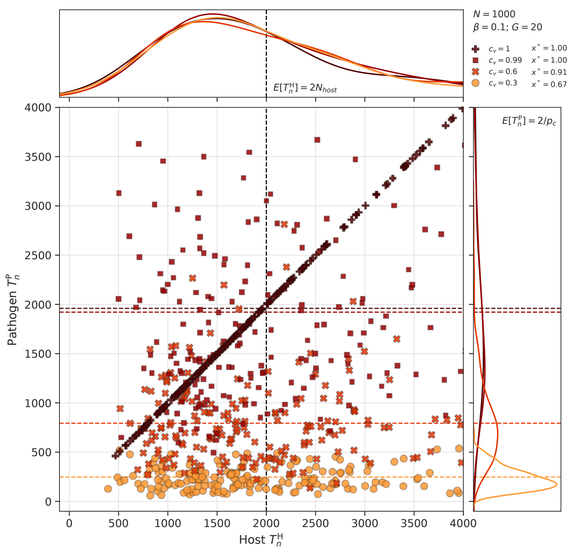
Stochastic, individual-based simulations with *SLiM*



Host and *pathogen* coalescent trees are simulated using *tree-sequence recording* in a *Two-Species* model!

Figure 4: Forward simulation of sochastic dynamics of the model. The proportion of infected at the end of each generation (solid line) fluctuates around the deterministic solution x^*. Following vertical transmission during host reproduction, the epidemic cycle within each host generation leads to oscillations.

Joint Host-Pathogen Genealogies



Under complete vertical transmission ($c_v = 1$) the height of the pathogen coalescent tree T_n matches that of the host.

Low levels of horizontal transmission strongly decouple the joint coalescence times of hosts and pathogens.

High horizontal relative to vertical transmission strongly reduces the pathogen T_n.

Figure 5: Simulated Host and Pathogen coalescence times of $n = 50$ infected pairs, for different vertical transmission probabilities c_v and resulting x^* values, over 200 independent replicates.

Key Take-Aways

We present a semi-discrete model of host and pathogen reproductive processes and their infection process, suitable for discrete generation hosts with epidemic cycles, such as annual plant pathogens.

Our model is a neutral model for studying the joint host and pathogen genealogies for which we derive analytical results and simulations of the embedded coalescence processes under horizontal and vertical transmission.

Future work and Open Questions

Reciprocal selection and coevolution. Frequency-dependent selection. Infection matrix. Fitness costs of infectivity and resistance.

Joint host-pathogen ARGs. Recombining host and pathogen genotypes. Co-infection of several strains.

References

[1] A. Tellier et al. “Theory of Host-Parasite Coevolution: From Ecology to Genomics” (2021).
[2] D. Ebert et al. “Host–parasite Co-Evolution and Its Genomic Signature”. *Nature Reviews Genetics* (2020).
[3] X. Didelot et al. “Ancestral Process for Infectious Disease Outbreaks With Superspreading”. *Journal of Theoretical Biology* (2025).
[4] B. C. Haller et al. “Slim 4: Multispecies Eco-Evolutionary Modeling”. *The American Naturalist* (2023).
[5] D. M. de Vienne et al. “Cospeciation Vs Host-shift Speciation: Methods for Testing, Evidence From Natural Associations and Relation To Coevolution”. *New Phytologist* (2013).

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